

Revised Roadmap: 3-4 Months to AI Developer

This is a comprehensive, topic-by-topic checklist with a timeline for a high-intensity, 3-4 month learning plan.

Month 1: The Foundations

- **Week 1: Python & Basics (Days 1-7)**
 - Python Syntax & Data Types (lists, dicts, tuples, sets).
 - Control Flow (**if/else**, loops) and Functions.
 - Object-Oriented Programming (OOP) concepts.
 - **Project:** Simple calculator or number guessing game.
- **Week 2: NumPy, Pandas & Git (Days 8-14)**
 - NumPy arrays, slicing, broadcasting, and vectorized operations.
 - Pandas DataFrames, data cleaning, filtering, and grouping.
 - Git/GitHub: Committing, pushing, and managing a code repository.
 - **Project:** Analyze and clean a simple dataset.
- **Week 3: Visualization & Foundational Math (Days 15-21)**
 - Data Visualization with Matplotlib and Seaborn.
 - Linear Algebra: Vectors, matrices, and dot products.
 - Calculus: The concepts of derivatives and gradients.
 - **Project:** Create visualizations to tell a story from your dataset.
- **Week 4: Probability, Stats & Portfolio Building (Days 22-28)**
 - Probability & Statistics: Descriptive stats (mean, median), Bayes' Theorem.
 - Finalize and document your foundational projects on GitHub.

Month 2: Core ML & The Deep Learning Bridge

- **Week 5: Core ML & Model Evaluation (Days 29-35)**
 - **ML Fundamentals:** Supervised vs. Unsupervised Learning, the ML workflow.
 - **Algorithms:** Linear/Logistic Regression, K-Nearest Neighbors (KNN).
 - **Metrics:** Accuracy, Precision, Recall, F1-Score, Confusion Matrix.
 - **Project:** Build a classifier on a classic dataset (e.g., Iris or Titanic).
- **Week 6: Unsupervised Learning & Deep Learning Intro (Days 36-42)**
 - **Unsupervised:** K-Means clustering, Principal Component Analysis (PCA).
 - **Deep Learning:** The Neuron, activation functions, and the concept of forward/backward propagation.
- **Week 7: Transformers & Embeddings (Days 43-49)**
 - **The Transformer Architecture:** Understand the intuition behind the attention mechanism.
 - **Text Embeddings:** Learn how to convert text into numerical vectors using pre-trained models from Hugging Face.
 - **Project:** Generate embeddings for a set of documents and perform a basic semantic search.
- **Week 8: LLMs & Prompt Engineering (Days 50-56)**
 - **LLM Concepts:** What LLMs are and how they differ from traditional models.
 - **API Usage:** Learn to use APIs from OpenAI or Hugging Face.

- **Prompt Engineering:** The art of creating effective prompts.
- **Project:** Build a simple chatbot using an LLM API.

Month 3: RAG & Fine-Tuning

- **Week 9: Retrieval-Augmented Generation (RAG) (Days 57-63)**
 - **Vector Databases:** The purpose and use of a vector database (e.g., ChromaDB, Pinecone).
 - **The RAG Pipeline:** The full process of document chunking, embedding, retrieval, and generation.
 - **Project:** Build a Q&A application over a custom PDF or set of documents using a library like LangChain or LlamaIndex.
- **Week 10: Fine-Tuning (Days 64-70)**
 - **Theory:** The difference between fine-tuning and prompt engineering, and when to use each.
 - **PEFT:** Learn about Parameter-Efficient Fine-Tuning (PEFT) and techniques like LoRA.
 - **Practice:** Fine-tune a pre-trained model on a specific task using the Hugging Face ecosystem.

Month 4: Specialization & Capstone Project

- **Week 11: Advanced Concepts & MLOps (Days 71-77)**
 - **Advanced RAG:** Learn multi-query retrieval and other advanced techniques.
 - **MLOps for LLMs:** Containerization with Docker and serving a model with a framework like FastAPI.
- **Weeks 12-16: The Capstone Project (Days 78-120)**
 - **Problem Definition:** Choose a problem to solve using RAG or fine-tuning that is more complex than your previous projects.
 - **Full Implementation:** Build the entire application from start to finish.
 - **Documentation:** Create a detailed README, explaining your project's architecture.
 - **Deployment:** Deploy a functional prototype to a cloud service to showcase your work.